SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: **Database Management Systems**
COURSE CODE: **CSA05**

COURSE OUTCOME COVERED

CO3: *Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity.* (BL2, BL3)

Activity Tool : MCQ Test (Low) Assessment
Tool Weightage: 30%

QUESTIONS		Total Marks: 50	
Q.No	**Question**	**Bloom's Level**	**Marks**
1	Which of the following is true about 1NF? (a) Allows multi-valued attributes (b) Eliminates partial dependencies (c) All attributes must be atomic (d) Eliminates transitive dependencies	BL2 – Understand	**5**
2	A relation is in 2NF if it is in 1NF and: (a) Has no transitive dependencies (b) Has no partial dependencies on primary key (c) Every attribute is a prime attribute (d) Has no multi-valued dependencies	BL2 – Understand	**5**
3	Which normal form deals with multi-valued dependencies? (a) 2NF (b) 3NF (c) BCNF (d) 4NF	BL2 – Understand	**5**
4	BCNF is a stricter version of: (a) 1NF (b) 2NF (c) 3NF (d) 4NF	BL2 – Understand	**5**
5	Functional dependency $X \rightarrow Y$ means: (a) Y determines X (b) X determines Y (c) X and Y are unrelated (d) Both determine each other	BL2 – Understand	**5**
6	Which of the following anomaly is NOT associated with unnormalized relations? (a) Insertion anomaly (b) Deletion anomaly (c) Update anomaly (d) Retrieval anomaly	BL2 – Understand	**5**
7	A superkey is: (a) A minimal set of attributes that uniquely identifies tuples (b) Any set of attributes that uniquely identifies tuples (c) An	BL2 – Understand	**5**

	attribute that depends on a non-prime attribute (d) A foreign key reference		
8	If R(A,B,C) has FD: A→B and B→C, then C is: (a) Partially dependent on A (b) Fully dependent on A (c) Transitively dependent on A (d) Independently determined	BL2 – Understand	**5**
9	5NF is also known as: (a) Domain-Key Normal Form (b) Boyce-Codd Normal Form (c) Project-Join Normal Form (d) Element Normal Form	BL2 – Understand	**5**
10	Lossless-join decomposition ensures: (a) No data is lost during decomposition (b) All FDs are preserved (c) The relation is in BCNF (d) No null values exist	BL2 – Understand	**5**

Code of Conduct:

I M.Devi Srilatha (192525236)*(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

M. Devi Srilatha

RUBRICS FOR CALCULATION:

MCQ					
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to problem-based questions	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

1.Which of the following is true about 1NF? (a) Allows multi-valued attributes (b) Eliminates partial dependencies (c) All attributes must be atomic (d) Eliminates transitive dependencies

A) (c) All attributes must be atomic

1NF – First Normal Form

1NF (First Normal Form) is the first step in database normalization. It ensures that the data in a table is organized properly and each field contains only **one atomic value**.

Main Rule of 1NF

All attributes must contain atomic (single) values.

This means:

- No multiple values in a single cell.
- No repeating groups of columns.
- Each row should be uniquely identifiable.

Example – Not in 1NF

StudentID	StudentName	PhoneNumbers
101	Ravi	9876, 8765
102	Priya	9123, 8234

Here, PhoneNumbers contains multiple values in one cell. Therefore, it is **not in 1NF**.

Convert to 1NF

StudentID	StudentName	PhoneNumber
101	Ravi	9876
101	Ravi	8765
102	Priya	9123
102	Priya	8234

Now each cell contains only **one value**, so the relation satisfies **1NF**.

Other Normal Forms

- **1NF:** Removes multi-valued/repeating attributes.
- **2NF:** Removes **partial dependencies**.
- **3NF:** Removes **transitive dependencies**.
- **BCNF:** Every determinant must be a candidate key.
- **4NF:** Removes non-trivial **multivalued dependencies**.
- **5NF:** Removes problematic **join dependencies**.

Therefore:

Correct answer: (c) All attributes must be atomic.

2. A relation is in 2NF if it is in 1NF and: (a) Has no transitive dependencies (b) Has no partial dependencies on primary key (c) Every attribute is a prime attribute (d) Has no multi-valued dependencies

A) (b) Has no partial dependencies on primary key

2NF – Second Normal Form

Correct Answer: (b) Has no partial dependencies on primary key ✅

Explanation

Second Normal Form (2NF) is a level of database normalization that removes **partial dependency** from a relation.

A relation is in **2NF** when:

1. It is already in **1NF**.
2. There is **no partial dependency** of a non-key attribute on a part of a composite primary key.

Partial Dependency

A partial dependency occurs when a **non-key attribute depends on only part of a composite primary key**, instead of the complete primary key.

Example

Consider:

ENROLLMENT(StudentID, CourseID, StudentName, CourseName, Grade)

Primary Key:

(StudentID, CourseID)

Functional dependencies:

StudentID → StudentName

CourseID → CourseName

(StudentID, CourseID) → Grade

Here:

- StudentName depends only on StudentID.
- CourseName depends only on CourseID.
- Grade depends on the complete (StudentID, CourseID).

Therefore, StudentName and CourseName have **partial dependencies**, so the relation is **not in 2NF**.

3. Which normal form deals with multi-valued dependencies? (a) 2NF
(b) 3NF (c) BCNF (d) 4NF

A) (d) 4NF

Fourth Normal Form (4NF) deals with **multivalued dependencies (MVDs)**.

A relation is in **4NF** when:

1. It is already in **BCNF**.
2. It has **no non-trivial multivalued dependencies** unless the determinant is a **superkey**.

Multivalued Dependency?

A multivalued dependency occurs when one attribute determines **multiple independent values** of another attribute.

Example

Consider:

STUDENT(StudentID, Hobby, Language)

Suppose a student can have:

- Multiple hobbies
- Multiple languages

and hobbies and languages are independent.

For example:

StudentID	Hobby	Language
101	Cricket	English
101	Cricket	Telugu
101	Music	English
101	Music	Telugu

This creates unnecessary combinations.

The multivalued dependencies are:

StudentID $\rightarrow\rightarrow$ Hobby

StudentID $\rightarrow\rightarrow$ Language

Decomposition into 4NF

Separate the independent multivalued attributes:

STUDENT_HOBBY(StudentID, Hobby)

STUDENT_LANGUAGE(StudentID, Language)

Now the multivalued dependencies are separated, and the relations satisfy **4NF**.

3. **BCNF is a stricter version of: (a) 1NF (b) 2NF (c) 3NF (d) 4NF**

   A) (c) 3NF

BCNF (Boyce-Codd Normal Form) is a **stricter version of 3NF**. Every relation that is in BCNF is also in 3NF, but a relation can be in 3NF and still violate BCNF.

Main Rule of BCNF

A relation is in **BCNF** if, for every non-trivial functional dependency:

$X \rightarrow Y$

X must be a superkey.

In simple words:

Every determinant must be a candidate key or a superkey.

Example

Consider:

TEACHER(TeacherID, Course, Room)

Suppose:

TeacherID → Course

Course → Room

If `Course` is not a superkey but determines `Room`, then the relation can violate BCNF.

To achieve BCNF, we decompose the relation into suitable smaller relations where every determinant is a superkey.

4. **Functional dependency X → Y means: (a) Y determines X (b) X determines Y (c) X and Y are unrelated (d) Both determine each other**

   A) (b) X determines Y

A **functional dependency (FD)** describes a relationship between two attributes in a database.

When we write:

$X \rightarrow Y$

it means:

The value of X uniquely determines the value of Y.

In other words, if two rows have the same value of **X**, they must have the same value of **Y**.

Example

Consider:

STUDENT(StudentID, StudentName, Age)

The functional dependency is:

StudentID → StudentName

This means **StudentID determines StudentName**.

For example:

StudentID	StudentName
101	Ravi
102	Priya
103	Arun

If we know:

StudentID = 101

we can determine:

StudentName = Ravi

But knowing StudentName does not necessarily determine StudentID, because two people could have the same name.

6. Which of the following anomaly is NOT associated with unnormalized relations?

(a) Insertion anomaly (b) Deletion anomaly (c) Update anomaly (d) Retrieval anomaly

A) (d) Retrieval anomaly

Unnormalized or poorly designed relations commonly suffer from three major anomalies:

1. Insertion Anomaly

Occurs when we cannot insert new information without adding unrelated information.

Example:

We cannot add a new course unless at least one student enrolls in it.

2. Deletion Anomaly

Occurs when deleting one record accidentally removes other important information.

Example:

Deleting the only student enrolled in a course may also delete the course information.

3. Update Anomaly

Occurs when the same information is stored in multiple places and must be updated everywhere.

Example:

If a teacher's name appears in 10 records, changing the name requires updating all 10 records.

Retrieval Anomaly

Retrieval anomaly is **not one of the standard anomalies associated with normalization**.

Normalization mainly addresses:

Insertion, Deletion, and Update anomalies.

7. A superkey is: (a) A minimal set of attributes that uniquely identifies tuples (b) Any set of attributes that uniquely identifies tuples (c) An attribute that depends on a non-prime attribute (d) A foreign key reference

A) (b) Any set of attributes that uniquely identifies tuples

Superkey

A **superkey** is a set of one or more attributes that can **uniquely identify every row (tuple)** in a relation.

The important point is that a superkey **does not have to be minimal**.

Example

Consider:

STUDENT(StudentID, StudentName, Email)

Suppose StudentID uniquely identifies each student.

Then these can be superkeys:

- {StudentID}
- {StudentID, StudentName}
- {StudentID, Email}
- {StudentID, StudentName, Email}

All of them uniquely identify a student.

However, **{StudentID} is a candidate key** because it is the **minimal** superkey.

Difference

Concept	Meaning
Superkey	Any set of attributes that uniquely identifies tuples
Candidate Key	Minimal superkey
Primary Key	Candidate key selected as the main key

8. If R(A,B,C) has FD: A→B and B→C, then C is: (a) Partially dependent on A (b) Fully dependent on A (c) Transitively dependent on A (d) Independently determined

A) (c) Transitively dependent on A

Given the relation:

R(A, B, C)

and functional dependencies:

A → B

B → C

We can derive:

A → B → C

Therefore, **C is transitively dependent on A**.

Transitive Dependency

A transitive dependency occurs when:

A → B

B → C

and therefore:

A → C

where **B is an intermediate attribute**.

Example

Consider:

STUDENT(StudentID, DepartmentID, DepartmentName)

Functional dependencies:

StudentID → DepartmentID

DepartmentID → DepartmentName

Therefore:

StudentID → DepartmentID → DepartmentName

So DepartmentName is **transitively dependent** on StudentID.

9. 5NF is also known as: (a) Domain-Key Normal Form (b)Boyce-Codd Normal Form (c) Project-Join Normal Form (d) Element Normal Form

A) (c) Project-Join Normal Form

5NF (Fifth Normal Form) is also called Project-Join Normal Form (PJ/NF).

It deals mainly with join dependencies. A relation is in 5NF when it cannot be further decomposed into smaller relations without losing information or introducing spurious tuples, except when the decomposition is implied by candidate keys.

Main Concept

5NF → Deals with Join Dependencies

For example, a relation may contain independent relationships between:

Student ↔ Course

Student ↔ Subject

Course ↔ Subject

If these relationships create unnecessary combinations, the relation may need to be decomposed into smaller relations.

10. Lossless-join decomposition ensures: (a) No data is lost during decomposition (b) All FDs are preserved (c) The relation is in BCNF(d) No null values exist

A) (a) No data is lost during decomposition

Lossless-Join Decomposition

A lossless-join decomposition means that when a relation is divided into smaller relations, we can join the smaller relations back together and obtain the original relation without losing information or generating incorrect (spurious) tuples.

Example

Suppose we have:

STUDENT(StudentID, StudentName, CourseID, CourseName)

We decompose it into:

STUDENT(StudentID, StudentName)

COURSE(CourseID, CourseName)

ENROLLMENT(StudentID, CourseID)

If these tables can be correctly joined to reconstruct the original information, the decomposition is **lossless**.

Important Point

Lossless join guarantees:

Original information can be recovered after decomposition.

It does not necessarily guarantee that all functional dependencies are preserved.